

Retail Pulse

End to End Retail Analytics Case Study

A two-year retail analytics investigation spanning 2023 to 2024.

Technology: PostgreSQL • SQL • Power BI/DAX • Figma

Data → Analysis → Insight → Business Implication → Recommendation

1. Project Overview

Retail Pulse is an end to end retail analytics project designed to evaluate sales performance, customer behavior, product contribution, and purchasing funnel efficiency across a two year retail dataset spanning 2023 2024.

The project was designed to go beyond descriptive reporting. Rather than simply presenting what happened, the analysis investigates the patterns behind business performance, identifies areas that warrant further investigation, and translates the findings into practical business opportunities.

The analysis combines PostgreSQL, SQL, Power BI/DAX, and Figma into a single workflow.

The resulting Power BI dashboard consists of five analytical pages:

- Executive Summary
- Sales Performance
- Customer Intelligence
- Product Intelligence
- Funnel Performance

Data → Analysis → Insight → Business Implication → Recommendation

2. Business Objective & Analytical Questions

The primary objective was to identify the key drivers of revenue performance and uncover opportunities to improve customer value, retention, product performance, pricing, and conversion efficiency.

The analysis was structured around six questions:

- How is the business performing, and what is contributing to revenue growth?
- Which customers and customer segments contribute the most value?
- Which products and categories drive revenue versus sales volume?
- How do pricing, purchasing activity, and product mix change across different periods?
- Where are customers being lost throughout the purchasing funnel?
- What actions could potentially improve revenue and commercial efficiency?

3. Data & Analytical Approach

Data

The analysis uses six related datasets:

- Customers
- Orders
- Order Items
- Products
- Events
- Regions

The datasets were initially assessed for missing values, inconsistencies, and data type issues before being loaded into PostgreSQL. No major cleaning intervention was required, allowing the project to focus primarily on analytical modeling and business interpretation.

Analytical workflow

Raw datasets → Data validation → PostgreSQL → SQL analysis → Analytical views → Power BI data model → DAX measures → Dashboard design → Interactive analysis → Insights & recommendations

PostgreSQL formed the analytical layer, where SQL was used to answer the core business questions and create reusable analytical views.

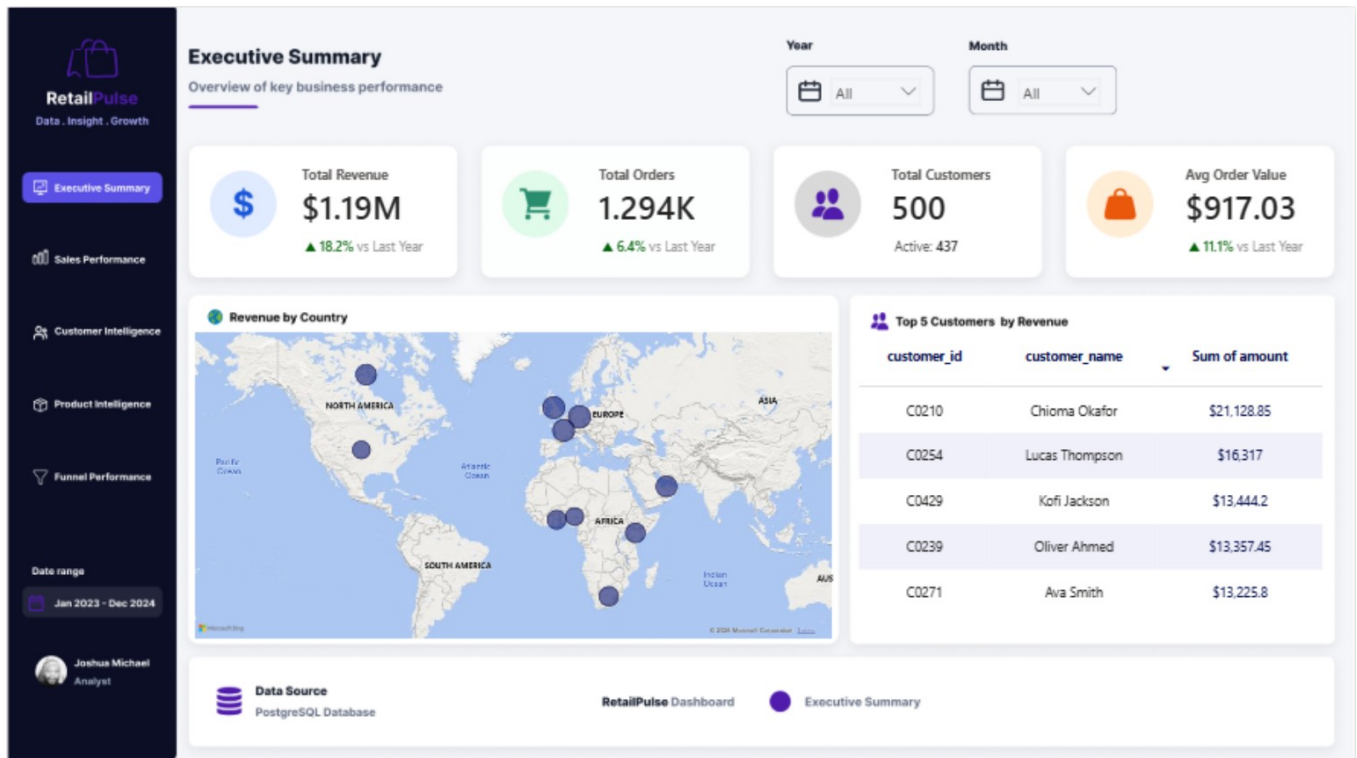
Key techniques included:

- Aggregation and conditional aggregation
- Common Table Expressions (CTEs)
- LAG() for period over period analysis
- ROW_NUMBER() for product ranking
- NTILE() for RFM scoring
- Cohort analysis
- RFM segmentation
- Funnel analysis
- Revenue per customer analysis
- Time series analysis
- Product and category analysis
- Seasonal price and volume analysis

Power BI was then used as the visualization and exploration layer, with DAX measures supporting the dashboard calculations.

Figma was used to plan and refine the dashboard's visual structure before implementation.

4.1. Executive Summary



Dashboard screenshot for the corresponding analytical page.

Purpose

The Executive Summary provides a high level view of business performance before moving into the deeper analyses. It focuses on revenue growth, order activity, customer participation, geographic contribution, and customer concentration.

Key Metrics

- Revenue
- Orders
- Average Order Value (AOV)
- Active Customers
- Revenue by Country
- Customer revenue concentration

Revenue increased by 18.2% year over year, while orders increased by 6.4% and AOV increased by 11.1%.

Revenue growth was supported by both increased purchasing activity and higher value per transaction, with the larger contribution coming from the increase in AOV. In other words, the business generated more revenue not only by processing more orders, but by generating substantially more revenue from each order. This indicates that revenue growth was driven more strongly by higher order value than by increased order volume.

The geographic analysis showed that Europe is the company's strongest regional market, contributing approximately 36.5% of total revenue.

However, revenue was relatively distributed at the individual country level, with Germany, the largest individual market, accounting for approximately 12.7%, while France and the UK each contributed around 12%. Revenue remains geographically diversified, with no single country accounting for more than 13% of total revenue.

This indicates regional concentration without strong dependence on a single country.

Customer revenue concentration was also relatively limited at the individual customer level. The five highest revenue customers collectively contributed approximately 6.5% of total revenue, indicating that the business was not disproportionately dependent on a small number of individual customers.

The later RFM analysis, however, revealed that customer value was still unevenly distributed across segments.

Business Implication

The business entered the deeper analysis from a position of revenue growth, but the headline numbers also raised several questions: What was driving the increase in transaction value? Was the growth consistent throughout the year? Which products and customers were responsible for the additional value? Where were opportunities being lost before conversion?

4.2. Sales Performance



Dashboard screenshot for the corresponding analytical page.

Purpose

The Sales Performance page examines how revenue evolved over time, focusing on growth, monthly fluctuations, stability, and potential seasonal patterns.

Key Metrics

- Monthly Revenue
- Month over Month Growth
- Year over Year Growth
- Running Revenue
- Median Monthly Revenue
- Revenue Volatility

Revenue increased from approximately \$543.9K in 2023 to \$642.8K in 2024, representing 18.2% growth.

Median monthly revenue increased from approximately \$38.52K to \$48.66K, an increase of about 26.3%.

At the same time, the coefficient of variation declined from approximately 45.1% to 29.6%, while average absolute month over month movement fell from 51.5% to 24.7%. Together, these measures indicate that 2024 was not only a stronger year but also a more consistent year.

Monthly patterns

Revenue fluctuated considerably throughout both years. July was the strongest month in 2023, while December became the strongest month in 2024. November and December also showed elevated performance across both years.

The repeated strength around July and the year end period suggests a potential seasonal component to demand. However, with only two years of observations, the analysis treats this as a seasonal pattern worth investigating rather than a conclusively established seasonal trend.

February 2023 anomaly

February 2023 was an unusually weak month and therefore represents another useful investigation point. The available data does not establish the cause. Potential areas to investigate include:

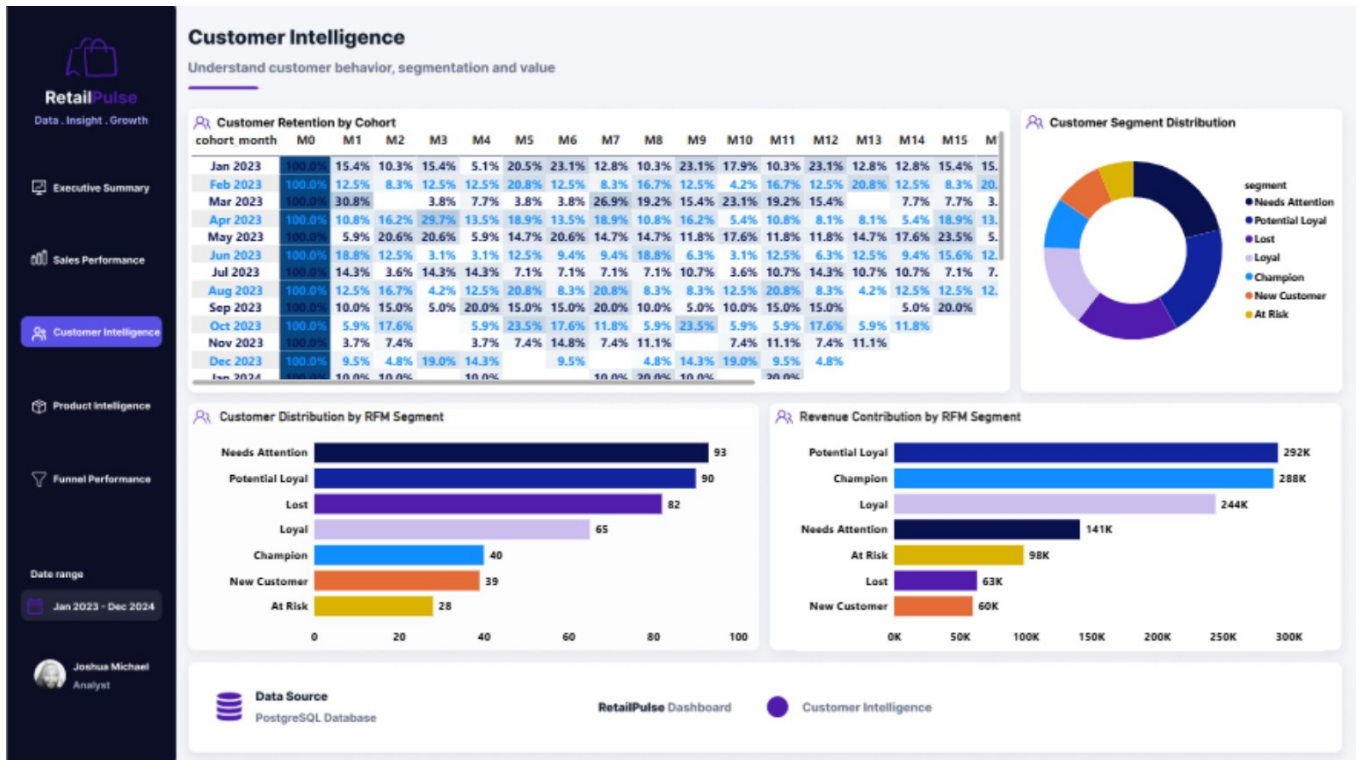
- Inventory or stock availability
- Changes in demand
- Pricing
- Order volume
- Average order value
- Marketing activity
- Competitive conditions
- Operational or supply chain issues

Business Implication

The business grew while becoming less volatile, suggesting improved revenue consistency and potentially greater predictability. The recurring stronger periods also create opportunities for the business to plan inventory, marketing, pricing, and promotional activity ahead of periods where demand has historically been stronger.

The sales analysis therefore established when performance changed. The Product and Customer analyses were then used to investigate what may have contributed to those changes.

4.3. Customer Intelligence



Dashboard screenshot for the corresponding analytical page.

Purpose

The Customer Intelligence page evaluates customer value, purchasing behavior, segmentation, and retention. The analysis combines RFM segmentation with cohort analysis to distinguish customers based on both their current value and their purchasing behavior over time.

Key Metrics

Customer Value

Customer value was highly unevenly distributed across segments.

The Champion segment represented approximately 9.15% of active customers and generated around \$288K in revenue, giving it the highest revenue per customer at approximately \$7.2K.

The Potential Loyal segment generated slightly more total revenue at approximately \$292K, but represented a substantially larger customer population. Its higher total contribution was therefore driven primarily by customer volume rather than higher individual value.

Customer volume and customer value are not proportional.

Champions and Loyal customers together represented approximately 24% of active customers while contributing approximately \$532K in revenue, demonstrating the economic importance of high value customer relationships.

At the other end of the distribution, Needs Attention represented the largest customer group but generated substantially less revenue per customer.

The analysis therefore shows value concentration at the segment level, even though the five largest individual customers represented only 6.5% of total revenue.

Retention Finding

The cohort analysis revealed a noticeable decline from Month 0 to Month 1 among cohorts with an observable second month period.

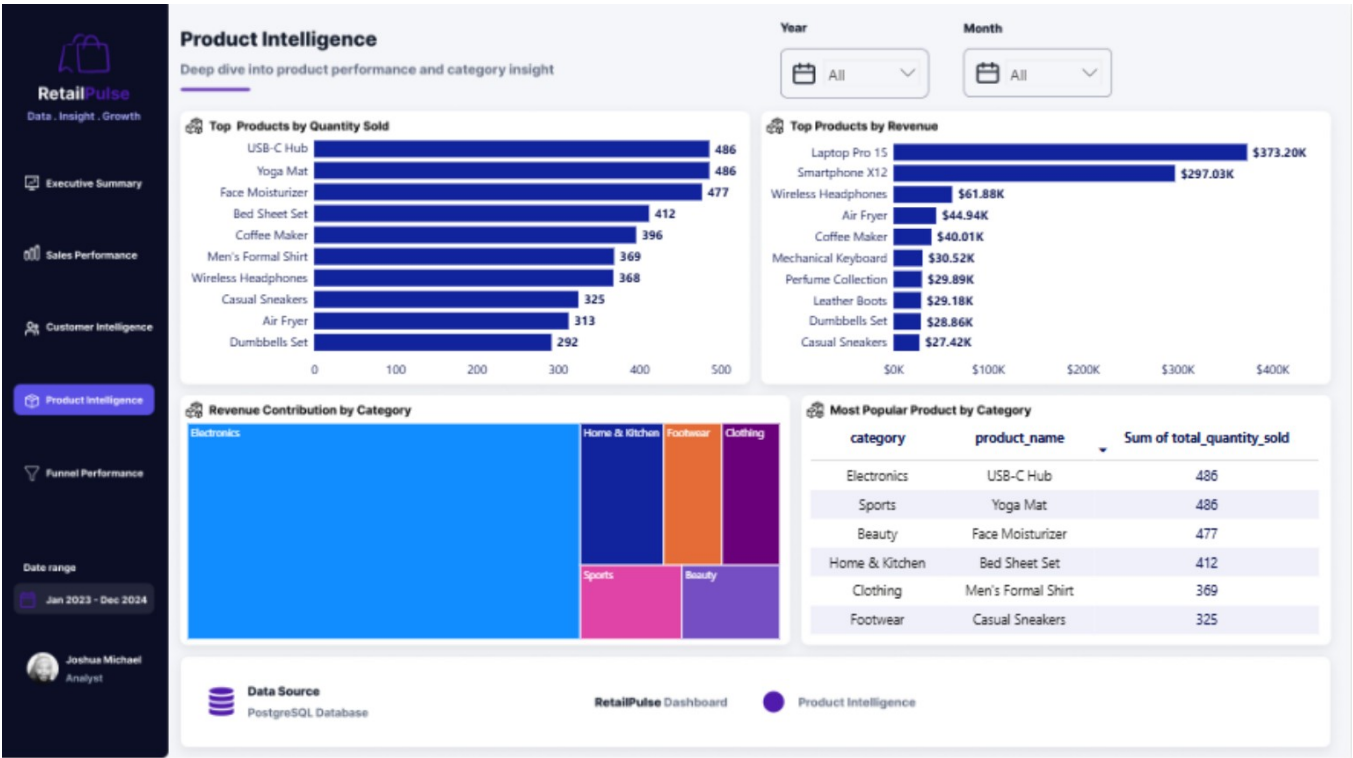
This indicates that the transition from the initial purchase to a subsequent purchase is an important point in the customer lifecycle. The finding does not establish that customers should necessarily purchase again within exactly 30 days, but it identifies the early post purchase period as a potential retention opportunity.

Business Implication

- Champions: protect through retention and loyalty strategies
- Loyal: encourage continued purchasing and value expansion
- Potential Loyal: develop toward higher value behavior
- At Risk: prioritize targeted win back activity
- New Customers: strengthen post purchase engagement
- Needs Attention: attempt reactivation before further inactivity

The analysis therefore suggests that an important growth opportunity lies not only in acquiring additional customers, but also in increasing the value and retention of customers already acquired.

4.4. Product Intelligence



Dashboard screenshot for the corresponding analytical page.

Purpose

The Product Intelligence page examines product and category contribution, distinguishing revenue performance from unit volume while investigating seasonal changes in pricing and purchasing behavior.

Key Metrics

Category Contribution

Electronics generated approximately \$787.1K, representing 66.3% of total revenue across the full period.

Its contribution increased further from 63.6% in 2023 to 68.6% in 2024, while category revenue increased by approximately 27.4%.

Overall quantity sold also increased by approximately 8.5% in 2024.

This shows that the business was not simply generating more revenue through higher prices; sales volume also increased.

At the same time, the category analysis demonstrates that unit volume and revenue contribution are different measures of product importance. Lower priced categories can generate substantial unit sales without contributing proportionally to revenue.

Business Implication

Electronics represents both a major revenue opportunity and a concentration point. Its strong performance makes it an important revenue engine, but increasing dependence on one category also creates exposure to changes in Electronics demand, pricing, inventory, or availability.

The appropriate response is therefore not simply to reduce Electronics' importance, but to protect its performance while identifying opportunities to strengthen other categories.

Seasonal Pricing & Purchasing Behavior

One of the most significant findings emerged from comparing purchasing activity across different periods.

Product prices followed a recurring seasonal pattern, with increases during June August and again during November December.

For example, the Laptop Pro 15 increased from its \$1,200 base price to approximately:

- \$1,380 during June August (+15%)
- \$1,680 during November December (+40%)

Comparing June August with March May 2023:

Metric	Change
Revenue	+60.2%
Orders	+32.4%
Quantity	+28.5%
AOV	+20.8%

The November December period showed an even larger difference compared with September October:

Metric	Change
Revenue	+124.7%
Orders	+18.9%
Quantity	+43.8%
AOV	+88.3%

What the data shows

- Higher purchasing activity
- Higher quantities
- Higher AOV
- Higher revenue
- Greater contribution from high value products

What this suggests

The seasonal revenue peaks cannot reasonably be attributed to price increases alone. The evidence suggests an interaction between seasonal demand + higher purchasing activity + higher prices + high value product mix.

This creates a seasonal pricing hypothesis: stronger demand periods may provide greater pricing flexibility. However, because the data is observational rather than experimental, it does not establish that the price increases caused the revenue growth or that customers were specifically less price sensitive.

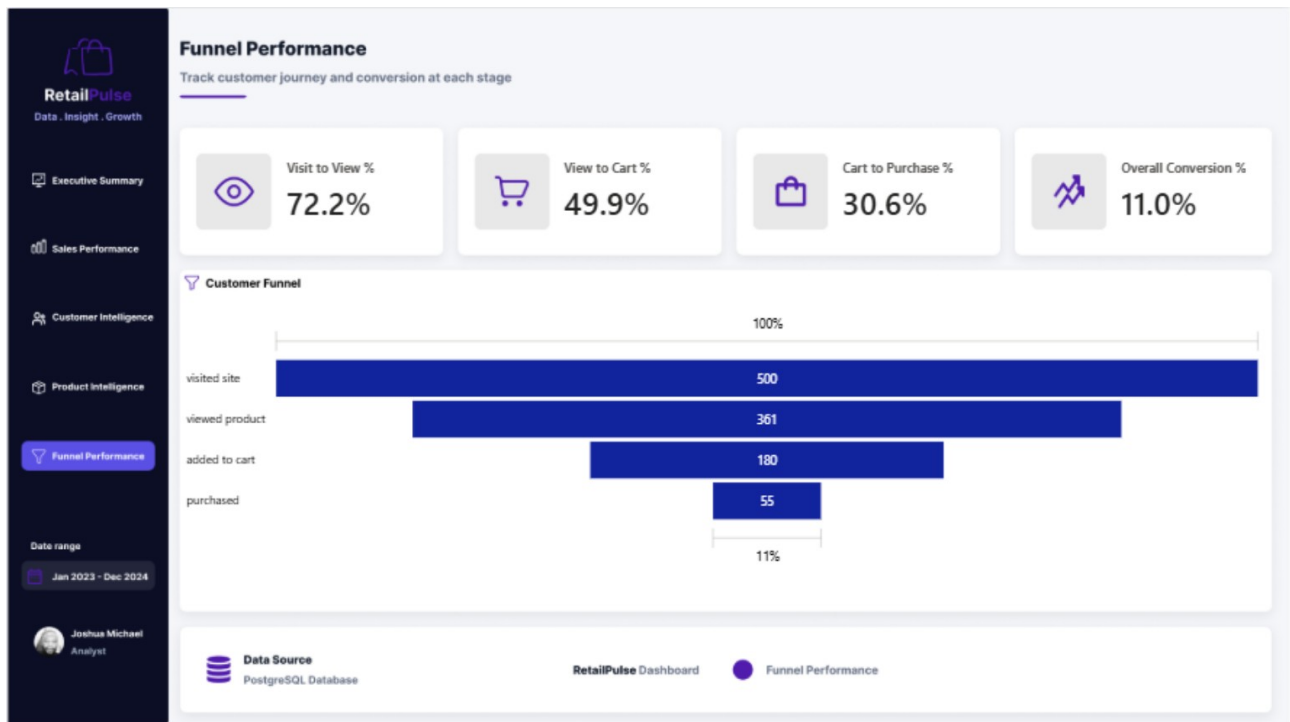
Business Implication

The business could investigate whether higher seasonal price points can be sustained without materially reducing demand. Before broad implementation, this should be validated through:

- Product level price elasticity analysis
- Margin analysis
- Controlled pricing experiments

- Conversion rate analysis at different price points

4.5. Funnel Performance



Purpose

The Funnel Performance page shows how visitors move from visiting the site to viewing products, adding products to their cart, and completing a purchase. The main goal is to see where customers are being lost.

Funnel Snapshot

Stage	Customers	Conversion
Visited site	500	100.0%
Viewed product	361	72.2%
Added to cart	180	49.9%
Purchased	55	11.0% overall

What the funnel shows

Strong initial engagement. 361 of 500 visitors viewed a product, giving a 72.2% visit to product view rate. This shows that most recorded visitors moved beyond the initial site visit and engaged with the product offering.

Strong product consideration. 180 of the 361 product viewers added an item to their cart. That is a 49.9% conversion rate, meaning roughly one in every two product viewers reached the cart stage.

The main leakage happens at checkout. 180 customers added products to their carts, but only 55 completed a purchase. This leaves 125 cart additions that did not become purchases, or 69.4% of cart additions.

Overall, the funnel converts 11.0% of site visitors into purchases. The number itself is best treated as the current result rather than labelled high or low without a relevant benchmark.

4.5. Funnel Performance, Continued

The business story

The funnel becomes more useful when viewed from top to bottom. The business is getting visitors to products and moving about half of product viewers into carts. The biggest loss happens after customers have already added a product to their cart.

This means the immediate opportunity is not simply to bring in more visitors. There is already a group of customers showing clear purchase interest. Understanding why they leave before completing checkout could improve purchases from the traffic the business already has.

What to investigate

Checkout costs

Check whether shipping, taxes, or other fees appear late and change the expected final price.

Payment experience

Review payment methods and payment failures to see whether customers are blocked at the final step.

Checkout steps

Check whether account creation, long forms, or too many steps are creating unnecessary friction.

Delivery and trust

Review delivery times, return information, and other factors that may affect the decision to complete the order.

Customer and product patterns

Compare abandonment by device, location, product category, and order value to find where the problem is most concentrated.

These are areas to investigate, not confirmed causes. The funnel tells us *where* customers are being lost, but the available data does not tell us *why* they leave.

Business recommendation

Prioritize checkout optimization and investigate cart abandonment before putting heavy emphasis on increasing traffic. The first step should be to identify the main source of checkout friction and then test targeted improvements.

There are 125 recorded cart additions that did not become purchases. If half were recovered, that would represent about 62 to 63 additional purchases. This is only an illustration of the available opportunity, not a prediction of what the business will recover.

Funnel conclusion

The strongest finding is the gap between cart creation and completed purchase. The business is getting customers far enough into the journey to show purchase interest, but 69.4% of cart additions do not reach checkout completion. Improving this stage could create additional purchases from existing traffic before the business invests heavily in acquiring more visitors.

5. The Retail Pulse Story

The business is growing

Revenue increased 18.2%, supported by both more orders and higher AOV.



Growth also became more consistent

2024 revenue volatility declined substantially, with the coefficient of variation falling from 45.1% to 29.6%.



Revenue is increasingly concentrated in high value products

Electronics contributed 66.3% of total revenue, increasing its share from 63.6% to 68.6%.



Customer value is also unevenly distributed

Champions represented only 9.15% of active customers but generated approximately \$288K, while larger segments generated similar totals through customer volume.



Seasonal periods create significant revenue opportunities

Higher seasonal prices occurred alongside stronger purchasing activity, higher quantity, and higher AOV.



But opportunities remain in retention and conversion

The cohort analysis showed an early decline in customer activity, while 69.4% of cart additions failed to become purchases.

Retail Pulse reveals a growing and increasingly consistent retail business, with strong contribution from high value products and customers, significant seasonal revenue opportunities, and clear opportunities to improve customer retention and checkout conversion.

6. Strategic Business Priorities

Priority	What the analysis suggests
Protect high value revenue	Electronics and high value customer segments contribute disproportionately to revenue
Optimize seasonal monetization	Strong seasonal periods show simultaneous increases in price, volume, AOV, and revenue
Improve customer retention	Cohort analysis highlights an early decline after the initial purchase
Reduce checkout leakage	69.4% of cart additions do not convert to purchases

6.1 Protect High Value Revenue

Protect the performance of Electronics and high value customer segments through appropriate availability, retention, and product strategies. At the same time, monitor category concentration and explore opportunities to strengthen other categories.

6.2 Optimize Seasonal Monetization

Evaluate seasonal pricing at product level using price, quantity, revenue, conversion, and margin together. The objective is not simply to increase prices, but to determine where seasonal demand provides sufficient pricing flexibility without materially reducing volume or profitability.

6.3 Improve Customer Retention

Develop segment specific engagement strategies, particularly around the transition from first purchase to subsequent purchasing. The effectiveness of retention initiatives should be measured through repeat purchase rate, revenue per customer, and movement between customer segments.

6.4 Reduce Checkout Leakage

Investigate the 69.4% cart abandonment rate before assuming that additional traffic is the primary growth opportunity. Further segmentation by product, geography, order value, device, payment method, and checkout stage could help identify the actual source of the leakage.

7. Analytical Certainty: What the Data Shows vs What Requires Testing

Observed

Electronics contributed 66.3% of revenue.

Observed

Product prices increased during particular seasonal periods.

Observed

Revenue, quantity, orders, and AOV also increased during those periods.

Reasonable inference

Seasonal demand and product mix appear to have contributed to the strength of those periods alongside higher prices.

Hypothesis requiring further analysis

Customers may have been less price sensitive during high demand periods.

Appropriate next step

Test price elasticity and, where possible, use controlled experiments to establish the effect of different price points.

This distinction prevents the analysis from turning correlation into causation and keeps the recommendations grounded in what the available data can actually support.

8. Limitations & Further Analysis

Two year observation period

Two years are sufficient to identify recurring patterns worth investigating, but not enough to establish long term seasonality with high confidence.

No cost or margin data

The project focuses primarily on revenue, orders, quantity, and customer behavior. High revenue therefore cannot automatically be interpreted as high profitability.

Observational pricing data

Prices and purchasing activity move together in the dataset, but the analysis cannot establish a causal price effect.

Limited funnel diagnostics

The funnel identifies where customers are lost but does not explain why.

Different cohort observation windows

Older cohorts have more observable months than newer cohorts, so retention comparisons need to account for the different observation periods.

RFM segmentation is dynamic

RFM segments describe customer behavior within the analysis period. Customers can move between segments as their behavior changes.

Further analysis

- Customer Lifetime Value analysis
- Churn prediction
- Segment migration analysis
- Product level price elasticity
- A/B pricing experiments
- Checkout/device/payment analysis
- Revenue forecasting
- Statistical testing of seasonal differences

9. Technical Implementation

Retail Pulse combines PostgreSQL, SQL, Power BI/DAX, and Figma.

SQL was used for the analytical layer, including complex transformations and reusable views for:

- Cohort analysis
- RFM analysis
- Funnel analysis
- Top product by category

Common Table Expressions were used to structure multi stage analyses into logical steps.

LAG() was used to compare current revenue with previous periods:

```
LAG(revenue) OVER ( ORDER BY month_start ) AS previous_month_revenue
```

RFM scoring used NTILE(5) to create relative customer groups:

```
(6 - NTILE(5) OVER ( ORDER BY recency_days ASC )) AS r_score
```

This approach ensured that more recent customers received higher recency scores.

Power BI was then used to build the interactive analytical layer through DAX measures, slicers, tooltips, time based analysis, and dashboard navigation.

Figma supported the visual planning and refinement of the dashboard before implementation.

10. Conclusion

Retail Pulse demonstrates an end to end approach to retail analytics that moves beyond simply presenting metrics.

The analysis found that revenue increased 18.2% year over year, while the business also became considerably more stable in 2024. Growth was supported by higher order volume and transaction value, while Electronics emerged as the dominant revenue category.

Customer analysis showed that value is unevenly distributed across segments, creating opportunities to protect high value relationships and improve the progression of customers toward higher value behavior.

Product analysis revealed strong seasonal patterns in both pricing and purchasing activity, creating a hypothesis around seasonal pricing flexibility that can be investigated further through elasticity analysis and controlled experimentation.

Finally, the funnel revealed a significant conversion opportunity, with 69.4% of cart additions failing to result in a purchase.

The resulting business story is therefore not simply that revenue increased.

It is that Retail Pulse identified how that growth is distributed, where it is concentrated, where additional value may be available, and where the customer journey is losing potential revenue.

Raw data → structured analysis → business insight → evidence based recommendations